CLINICAL · NECK, THYROID & SALIVARY

Radiotherapy and chemotherapy for head and neck cancer

Most patients with head and neck cancer receive radiotherapy at some point: alone for early tumours, with cisplatin for advanced ones, after surgery when the pathology shows risk, or for palliation. The surgeon needs to know enough radiation biology and chemotherapy to choose the right treatment, prepare the patient (teeth first), recognise the toxicities that come back to the ENT clinic, and know when a dry mouth, a stiff jaw or an exposed bone is the radiotherapy and when it is the cancer returning.

AT A GLANCE

- Conventional 2 Gy a day; 70 Gy for gross disease.
- Positive margins or extranodal extension: add cisplatin.
- Cisplatin is ototoxic: baseline audiogram.
- Osteoradionecrosis: prevent by dental care before treatment.

1 CHAPTER ONE Radiation basics

Physics and biology.

1.1 Radiation and dose

Cancer is treated with *ionising* radiation of short wavelength and high energy: photons, either X-rays from machines or gamma rays from isotopes such as cobalt-60, and particles such as electrons, protons, neutrons and carbon ions. Dose is measured in **grays**; 1 Gy is 1 joule absorbed per kilogram and equals 100 rad.

Most treatment uses megavoltage photons of 4 to 25 MV from a *linear accelerator*, which spare the skin because the maximum dose builds up below the surface. **Electrons** deposit their dose near the surface and fall off sharply, which suits the skin, lip and superficial nodes overlying the spinal cord. **Protons** deposit most of their energy at a set depth, the *Bragg peak*, with almost no exit dose, which suits the skull base, orbit and brainstem, re-irradiation and children. **Neutrons and carbon**

ions have a high linear energy transfer and a role in salivary and radioresistant tumours.

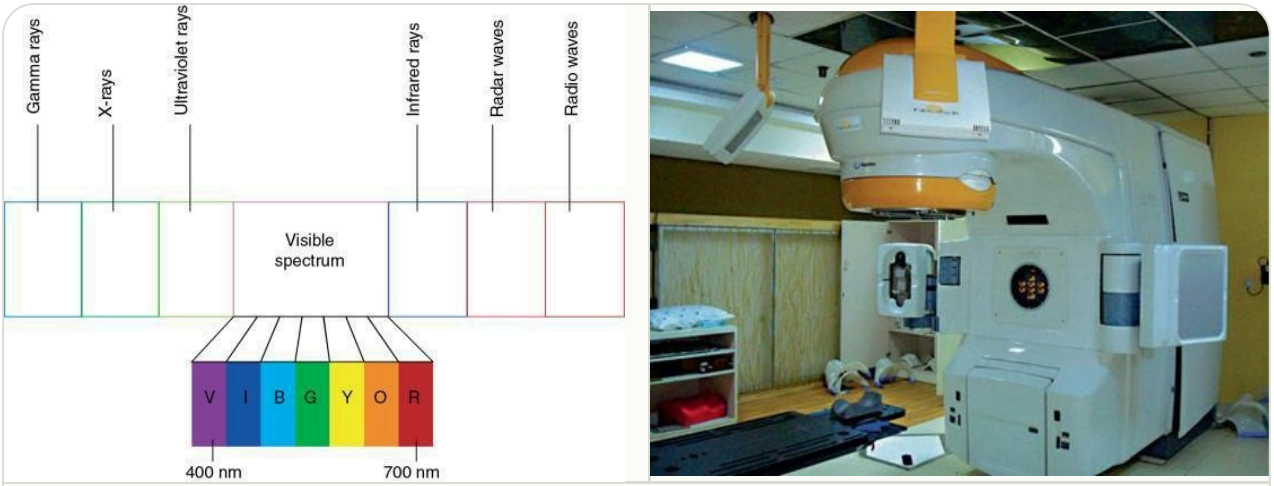


FIGURE 1 First, the electromagnetic spectrum. Second, a modern linear accelerator. Dhingra, figs 73.1 and 73.2

1.2 How radiation kills

Radiation damages DNA, mostly indirectly, by ionising water into free radicals that cause *double-strand breaks*; cells then die when they try to divide, or by apoptosis. Oxygen "fixes" the damage, so hypoxic cells are two to three times more resistant. That is why anaemia is corrected to a haemoglobin of about 12 g/dL, smoking is stopped, and hypoxic sensitisers such as nimorazole have been tried.

Fractionation works because of the **four R's**. Between fractions, normal tissues *repair* sublethal damage better than tumour cells; tumour cells *reoxygenate*; and cells *redistribute* into sensitive phases of the cycle. Against this, tumour *repopulation* accelerates after about four weeks, which is why gaps in treatment are harmful and overall treatment time matters.

The biological effect depends on dose per fraction as well as total dose, as described by the linear-quadratic model, $BED = nd(1 + d/(\alpha/\beta))$. Late-reacting tissues, with a low α/β of about 3, are spared by small fractions, whereas tumours and acutely reacting tissues, with an α/β of about 10, are not.

1.3 Fractionation

Conventional fractionation gives 1.8 to 2 Gy a day, five days a week. About 50 Gy sterilises microscopic disease such as the elective neck, 60 Gy is given after surgery, 66 Gy for close or positive margins, and 70 Gy in 35 fractions over seven weeks for gross tumour.

Hyperfractionation gives smaller fractions of 1.1 to 1.2 Gy twice daily to a higher total of 74 to 80 Gy, improving local control with fewer late effects but more acute toxicity. **Accelerated fractionation**, with six fractions a week or a concomitant boost, shortens overall time to counter repopulation, and helps mainly patients treated without chemotherapy. **Hypofractionation**, with fractions of 2.5 Gy or more, is used for palliation, frail patients and stereotactic treatment.

2 CHAPTER TWO Delivery

From mask to beam.

2.1 Planning

1. **Immobilisation:** a thermoplastic mask, with a mouth bite to displace the tongue.
2. **Planning CT,** fused with MRI or PET.
3. **Contouring:** the gross tumour volume (GTV); the clinical target volume (CTV), adding microscopic spread and nodal levels; the planning target volume (PTV), a margin for set-up error; and the *organs at risk*: spinal cord, brainstem, parotids, optic apparatus, cochlea, larynx, mandible and constrictors.
4. **Dose calculation** on a planning system.
5. **Daily image verification.**

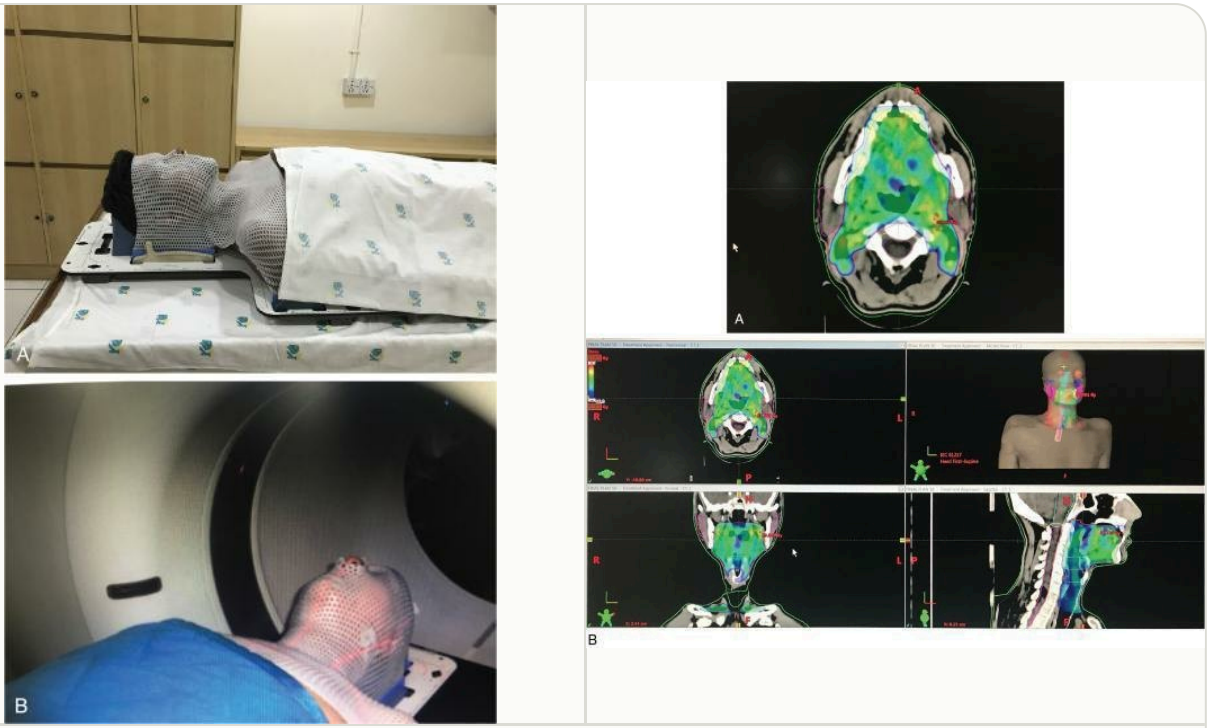


FIGURE 2 First, a thermoplastic immobilisation mask. Second, contouring of target volumes and organs at risk on the planning scan.

Dhingra, figs 73.3 and 73.4

2.2 Techniques

Conventional two-dimensional treatment with parallel opposed lateral fields is now historical, replaced first by CT-based **three-dimensional conformal radiotherapy** with several shaped beams and then by **intensity-modulated radiotherapy (IMRT)**, the current standard. IMRT uses inverse planning with many beams or arcs (VMAT), whose intensity is modulated by moving multileaf collimators, to "curve" the dose around the spinal cord and parotids and to give different doses to different targets at the same time, the *simultaneous integrated boost*. *Parotid-sparing IMRT* reduces permanent xerostomia.

Image-guided radiotherapy images the patient on the machine each day with cone-beam CT, correcting set-up errors and allowing smaller margins, and **adaptive radiotherapy** re-plans during treatment as the tumour shrinks and the patient loses weight. **Stereotactic radiosurgery and body radiotherapy** deliver very precise high doses in one to five sessions, for vestibular schwannoma (12 to 13 Gy in one session), paraganglioma, small recurrences and re-irradiation.

2.3 Brachytherapy

Brachytherapy places a sealed source in or against the tumour, interstitially or within a cavity, giving a high local dose that falls off rapidly. Iridium-192 is used with high-dose-rate remote afterloading, replacing caesium-137, radium and gold seeds. It treats cancers of the lip, oral tongue and floor of mouth and the nasopharynx, as a boost or for recurrence, and boosts external beam treatment.



FIGURE 3 First, interstitial brachytherapy needles for cancer of the lip. Second, brachytherapy of the tongue (A–C).
Dhingra, figs 73.5 and 73.6

3 CHAPTER THREE Radiotherapy in practice

When and why.

3.1 Roles

Definitive radiotherapy preserves the organ. Alone, it treats early T1 to T2 N0 tumours of the larynx (with control around 90% for T1 glottic cancer), oropharynx, nasopharynx and hypopharynx; with concurrent chemotherapy, it treats advanced tumours.

Postoperative radiotherapy starts four to six weeks after surgery, and the whole interval from surgery to the end of radiotherapy should be under about 11 weeks. It is indicated for close (under 5 mm) or positive margins, extranodal extension, two or more positive nodes or a node over 3 cm, pT3 and pT4 tumours, perineural or lymphovascular invasion, nodes in levels IV or V from oral or oropharyngeal cancer, and poor differentiation. *Positive margins and extranodal extension* are the indications for adding *concurrent cisplatin*.

Preoperative radiotherapy is now rarely used, because it increases wound complications and fistulae, and **palliative radiotherapy** uses short courses for pain, bleeding, fungation and obstruction.

PRE- VERSUS POSTOPERATIVE RADIOTHERAPY

PREOPERATIVE	POSTOPERATIVE
Shrinks borderline tumours and sterilises the edges	Surgery done on unirradiated tissue: easier, better healing
Oxygenated, undisturbed tumour bed	Dose tailored to pathology (margins, nodes); a higher dose possible
Smaller fields	Larger fields; scarred, hypoxic tumour bed
More fistulae, flap failures, carotid blow-out	Delay from surgical complications allows regrowth

3.2 Site by site

In the **larynx**, early glottic cancer is treated with radiotherapy alone or transoral laser surgery, and advanced cancer with concurrent cisplatin chemoradiation for larynx preservation, as established by RTOG 91-11; total laryngectomy is reserved for T4a disease with cartilage destruction or a non-functioning larynx. In the **hypopharynx**, larynx preservation is possible but less often successful.

In the **nasopharynx**, radiotherapy is the primary treatment, covering levels II and V and the retropharyngeal nodes; T1N0 disease is treated with radiotherapy alone and more advanced disease with cisplatin chemoradiation, often with induction or adjuvant chemotherapy, and plasma EBV DNA guides follow-up. In the **oropharynx**, radiotherapy or chemoradiation competes with transoral surgery followed by adjuvant treatment. HPV-positive disease has a much better outlook, with three-

year survival of 82% against 57% in RTOG 0129, but de-escalation is done only within trials.

Oral cavity cancer is primarily surgical, with adjuvant radiotherapy according to the pathology. **Salivary cancers** are operated on and then irradiated for high grade, positive margins, nodes or perineural spread; for adenoid cystic carcinoma the named nerve is covered to the skull base.

After chemoradiation, the **neck** is assessed with a PET-CT at 12 weeks, and dissection is done only for PET-positive residual disease, as shown by the PET-NECK trial. For **recurrence**, salvage surgery is preferred; re-irradiation, often hyperfractionated with chemotherapy or delivered by SBRT or protons, cures a minority at a high risk of toxicity, and the best candidates have a second primary many years after the first treatment.

TYPICAL DOSES

SETTING	DOSE
Elective neck (microscopic disease)	50 to 54 Gy
Postoperative, clear margins	60 Gy in 30 fractions
Postoperative, close or positive margins, extranodal extension	66 Gy in 33 fractions
Gross disease (definitive)	70 Gy in 35 fractions over 7 weeks
Early glottic cancer (T1)	63 Gy in 28 fractions, or 66 Gy in 33
Vestibular schwannoma (radiosurgery)	12 to 13 Gy single fraction

4 CHAPTER FOUR Chemotherapy and drugs

Partners to radiation.

4.1 Goals

Chemotherapy is given to sensitise the tumour to radiation and so improve locoregional control, to treat micrometastases, to preserve organs, to palliate recurrent or metastatic disease, and to debulk tumours before surgery or radiotherapy. On its own, it does not cure squamous carcinoma of the head and neck.

4.2 How it is given

Concurrent chemoradiation is the standard for locally advanced disease. The MACH-NC meta-analysis showed an absolute survival gain of about 6.5% at five years, all of it from concurrent treatment. *Cisplatin*, at 100 mg/m² every three weeks or 40 mg/m² weekly, is the drug of choice, and adding other drugs adds toxicity rather than cure.

Induction chemotherapy with *TPF* (docetaxel, cisplatin and 5-FU), which is better than PF, is used to select patients for larynx preservation and for bulky nodal disease, but it has not consistently beaten chemoradiation alone (DeCIDE, PARADIGM). **Adjuvant chemotherapy** after definitive treatment brings no benefit, except in nasopharyngeal cancer. **Postoperative chemoradiation** is given for positive margins and extranodal extension, on the evidence of the EORTC 22931 and RTOG 9501 trials.

4.3 Targeted therapy and immunotherapy

Cetuximab is a monoclonal antibody against EGFR, which is overexpressed in over 90% of these tumours. Combined with radiotherapy it improved survival over radiotherapy alone in the Bonner trial, at the cost of an acneiform rash and infusion reactions, and it is now used for patients unfit for cisplatin. With platinum and 5-FU it was the first-line treatment for metastatic disease in the EXTREME regimen.

The **PD-1 checkpoint inhibitors** nivolumab and pembrolizumab improved survival in recurrent or metastatic disease after platinum (CheckMate 141, KEYNOTE-040), and pembrolizumab, alone for PD-L1-positive tumours or with platinum and 5-FU, is now first-line treatment for recurrent or metastatic disease (KEYNOTE-048). Their immune-related side effects include thyroiditis, pneumonitis, colitis, hepatitis, adrenal insufficiency and rash.

CETUXIMAB IS NOT A SUBSTITUTE FOR CISPLATIN

In HPV-positive oropharyngeal cancer, radiotherapy with cetuximab gave worse survival and control than radiotherapy with cisplatin, without less toxicity (RTOG 1016, De-ESCALaTE HPV). Cisplatin remains the standard partner. Cetuximab is reserved for patients who cannot receive platinum.

DRUGS AND THEIR CHARACTERISTIC TOXICITIES

DRUG	USE	KEY TOXICITY
Cisplatin	Concurrent, induction, palliative	Nephrotoxicity (hydrate), high-frequency sensorineural hearing loss and tinnitus, neuropathy, severe nausea, marrow
Carboplatin	Cisplatin-unfit, weekly concurrent	Marrow (thrombocytopenia)
5-Fluorouracil	Induction (PF, TPF), palliative	Mucositis, diarrhoea, hand–foot syndrome, coronary spasm
Docetaxel, paclitaxel	Induction (TPF), palliative	Neutropenia, neuropathy, alopecia, hypersensitivity
Methotrexate	Palliative	Mucositis, marrow, renal and liver toxicity
Bleomycin	Formerly squamous and lymphoma	Pulmonary fibrosis, fever
Doxorubicin	Sarcomas, salivary cancers	Cardiomyopathy
Cetuximab	With radiotherapy if cisplatin-unfit; palliative	Acneiform rash, infusion reaction, hypomagnesaemia
Pembrolizumab, nivolumab	Recurrent and metastatic disease	Immune-related: thyroid, lung, gut, liver, adrenal

BEFORE CISPLATIN

- **Baseline audiogram:** cisplatin ototoxicity is cumulative and permanent, starting at 4 to 8 kHz.
- **Renal function:** creatinine clearance.
- **Blood count and electrolytes, including magnesium.**
- **Fitness for hydration:** check cardiac function.

Children and patients with existing hearing loss deserve monitoring during treatment. Sodium thiosulfate protects hearing in children.

5 CHAPTER FIVE Toxicity

Early and late.

5.1 Acute effects

Acute effects appear from the second week and settle within weeks of finishing. **Mucositis** causes pain, odynophagia and weight loss, and is managed with salt and bicarbonate mouth rinses, analgesia including opioids, treatment of candida, and nutritional support from a dietitian with a nasogastric tube or gastrostomy when needed. The **skin** reddens and then desquamates, dry and later moist, and is washed gently without rubbing or unprescribed creams. Taste is lost, saliva becomes sticky, the larynx swells, fatigue sets in, and chemotherapy suppresses the marrow.

5.2 Late effects

Late effects are permanent and depend on dose per fraction and total dose. **Xerostomia** follows a mean parotid dose above about 26 Gy and brings caries, candida, dysphagia and taste loss; it is reduced by parotid-sparing IMRT and treated with pilocarpine, saliva substitutes and, prophylactically, submandibular gland transfer. **Radiation caries** is prevented by daily fluoride trays and lifelong dental care, and **osteoradionecrosis** is described below.

Fibrosis of the soft tissues causes **trismus**, managed with jaw exercises and devices, and fibrosis of the constrictors causes **dysphagia**, stricture and aspiration, which is why swallowing exercises continue throughout treatment ("use it or lose it"). The **larynx** may develop oedema, chondronecrosis and a fixed cord. **Hypothyroidism** affects up to half of patients after neck radiotherapy, so TSH is checked every 6 to 12 months.

The **ear** suffers serous otitis media, sensorineural hearing loss that is worse with cisplatin, and radionecrosis of the canal and temporal bone. The **eye and nervous system** are at risk of cataract, retinopathy, *optic neuropathy*, brainstem and temporal lobe necrosis, and *radiation myelitis*, which is avoided by keeping the cord dose under about 45 to 50 Gy; Lhermitte's sign is an early, reversible form. Radiation accelerates **carotid atherosclerosis** and stroke, sets the scene for *carotid blow-out* after salvage surgery, and occasionally causes **second malignancies** such as sarcomas and thyroid cancer.

EARLY AND LATE COMPLICATIONS OF RADIOTHERAPY

EARLY	LATE
Mucositis, odynophagia	Permanent xerostomia, radiation caries
Skin erythema and desquamation	Skin atrophy, fibrosis, telangiectasia
Loss of taste, thick saliva	Osteoradionecrosis, chondronecrosis
Laryngeal oedema	Trismus, dysphagia, stricture
Candidiasis	Hypothyroidism, hypopituitarism
Marrow suppression	Serous otitis, sensorineural loss
Fatigue, nausea	Cataract, optic neuropathy, myelitis, brain necrosis; second cancers

5.3 Osteoradionecrosis (ORN)

Osteoradionecrosis is exposed, devitalised irradiated bone that fails to heal over three months in the absence of tumour. It almost always affects the posterior body of the *mandible*, where radiation leaves the bone hypovascular, hypocellular and hypoxic, with radiation-induced fibrosis. The risk rises with doses above 60 Gy,

dental extraction after radiotherapy, poor dentition, surgery on irradiated bone, smoking and alcohol.

It presents with pain and exposed bone, trismus, an orocutaneous fistula or a pathological fracture, and recurrence must always be excluded by biopsy and imaging. Prevention is the key: a dental assessment *before* radiotherapy, extraction of unsalvageable teeth at least two weeks before starting, fluoride, and atraumatic dentistry afterwards. Established disease is treated conservatively with irrigation, antibiotics and removal of loose sequestra; medically with pentoxifylline and tocopherol, with or without clodronate ("PENTOCLO"), and hyperbaric oxygen in selected cases; and, when advanced, by resection and reconstruction with vascularised bone such as a fibula free flap.

OPERATIVE GUIDES IN THEATRE 2 operations

Segmental mandibulectomy, and microvascular reconstruction.

Segmental Mandibulectomy
Segmental mandibulectomy

Microvascular Reconstruction of the Head and Neck
Free flap reconstruction

5.4 Care during treatment

Before treatment the mouth is cleared of unsalvageable teeth, smoking is stopped, anaemia is corrected, and the dietitian, speech and swallowing therapist and nurse specialist see the patient, with a prophylactic gastrostomy for selected high-risk patients. Each week the weight, blood count (and renal function with cisplatin), mucositis, skin care and analgesia are reviewed, with psychological support and help to stop drinking. Admission is needed for dehydration, neutropenic sepsis, aspiration or uncontrolled pain.

6 CHAPTER SIX Other tumours treated with radiation

Beyond squamous carcinoma.

6.1 Benign and unusual tumours

Stereotactic radiosurgery controls more than 95% of small and residual **vestibular schwannomas**, and radiotherapy or radiosurgery halts the growth of about 90% of **paragangliomas**. Unresectable intracranial residual **juvenile angiofibroma** is irradiated and regresses slowly over years. Radiotherapy is central in **sinonasal NK/T-cell lymphoma**, combined with asparaginase-based chemotherapy, and is part of combined treatment for **lymphoma of Waldeyer's ring** and chloroma. **Chordoma and chondrosarcoma** of the skull base receive high-dose proton therapy after maximal safe resection, and **olfactory neuroblastoma** receives postoperative radiotherapy after craniofacial or endoscopic resection.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The gray;** the four R's; why fractionate; why gaps and anaemia are bad.
- **IMRT and IGRT;** what parotid sparing achieves; proton Bragg peak.
- **Indications for postoperative radiotherapy,** and when to add cisplatin (margins, extranodal extension).
- **Chemoradiation:** cisplatin and its ototoxicity; the roles of TPF induction, cetuximab and immunotherapy.
- **Early and late toxicities;** osteoradionecrosis and its prevention.
- **PET-CT at 12 weeks** to decide on neck dissection after chemoradiation.

7.2 Common pitfalls

COMMON PITFALLS

- Starting radiotherapy before the teeth are assessed.
- Extracting teeth in an irradiated mandible without precautions.
- Long breaks in radiotherapy.
- Starting postoperative radiotherapy more than six weeks after surgery.
- Forgetting the TSH check and audiometry in follow-up.

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IMAGING

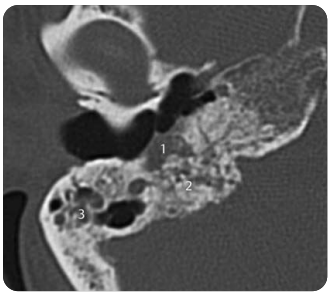
On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · BONE DISEASE

Osteoradionecrosis

- Stasis of secretions and infections are frequently the result of eustachian tube dysfunction due to mucosal swelling and loss of ciliary function. Chronic infections may require complete surgical removal of the affected bone and long-term antibiotic treatment.

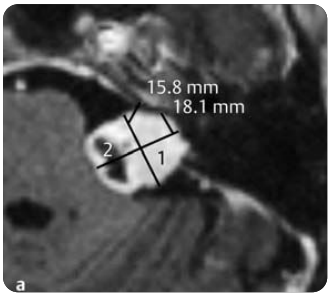


CT, axial. After radiotherapy for a residual glomus (1), the petrosal bone has a brittle appearance with signs of bony changes, fibrosis, and necrotize…

INNER EAR & IAC · AFTER SURGERY

Effects of Radiotherapy on CPA Schwannoma

- After stereotactic radiosurgery or fractionated radiotherapy, schwannomas often swell transiently for 6–18 months (pseudoprogression) and lose central enhancement before shrinking.
- Do not call failure on early growth alone; follow with serial MRI, and consider treatment failure only with sustained growth over about 2–3 years.
- Late effects to look for: hearing loss, trigeminal or facial symptoms, hydrocephalus and, rarely, radiation-induced malignancy.



MRI, T1-weighted with gadolinium enhancement, axial. In an attempt to preserve residual hearing, the patient opted for fractionated stereotactic…